An overlap bound between two unique crossings

Human Compression project: L044 continuation

Throughout $G = \langle \rho_0, \ldots, \rho_{r-1} \rangle \leq A_n$, $n \geq 12$, is transitive with a connected string C-marking, and every maximal omission G_k is intransitive. All actions are the given faithful permutation actions. An eligible edge of colour k crosses distinct G_k-orbits.

Lemma 1 (The outside orbit alone)**.** *Suppose the unique eligible i-edge joins $a \in A$ to $b \in B$, the two G_i-orbits, with $|B| \geq 2$, and $G_{<i}$ fixes B pointwise. If $h > i$ and ρ_h acts nontrivially on A, then*

$$|A| \geq i + h + 1. \tag{1}$$

For $i > 0$ the inequality is strict unless $h = i+1$. The conclusion uses no transitivity assumption on a final tail.

Proof. For $i = 0$, the accepted fixed-prefix/no-gap lemma supplies a simple path in A from a including a final h-edge, with at least h edges. Hence $|A| \geq h + 1$.

Let $i > 0$ and put $L = G_{<i}$. If L is transitive on A, every generator of index at least $i + 2$ fixes a and centralizes L, so fixes A pointwise. Thus $h = i + 1$. The accepted transitive-side bound gives $|A| \geq 2(i + 1)$, exactly (1).

Suppose L is intransitive on A. Its commuting-orbit normalization gives faithful equal-sized orbits of size $k \geq i + 1$; ρ_{i+1} exchanges the anchor orbit $\beta = a^L$ with another orbit β'. Put $d = h - i$. For $d = 1$, the two orbits give $|A| \geq 2k \geq 2i+2$. For $d = 2$, the first active orbit for ρ_h is β'. If it is moved to another orbit, there are at least three orbits and $|A| \geq 3k \geq 3i+3 > 2i+3$. If it is preserved nontrivially, the accepted internal-block lemma gives $A = \beta \cup \beta'$ and $i \leq k/2-1$. Consequently $|A| = 2k \geq 4i + 4 > 2i + 3$. The internal-block lemma's nontrivial-B hypothesis is explicit here.

For $d \geq 3$, the accepted active-orbit distance lemma gives at least d orbits. Therefore

$$|A| \geq d(i + 1) = 2i + d + i(d - 2) \geq 2i + d + 1 = i + h + 1.$$

Equality is possible only for $i = 1, d = 3, k = 2$ with exactly three L-orbits. But then $L = \langle \rho_0 \rangle$, and ρ_0 is a transposition on each of those three pairs and fixes B. It is odd, contrary to $G \leq A_n$. Thus this case is strict. □

Lemma 2 (Three regions and the middle path)**.** *Let $i < j$ be colours with unique eligible edges. Suppose the j-edge lies in the G_i-orbit B and the i-edge lies in the other side of the j-cut. Write the two edges as $a -_i b$ and $c -_j d$. Deleting them from the full permutation graph gives three regions A, M, D, with $a \in A$, $b, c \in M$, $d \in D$. If $j \geq i + 2$, then*

$$|M| \geq j - i. \tag{2}$$

Proof. A unique eligible edge is a bridge of the full permutation graph: all other edges preserve the two omission orbits, and that edge is their only connection. The full graph is connected by transitivity. Removing the two distinct bridges therefore gives exactly three connected components in the stated order. Here A is the first G_i-orbit and D is the G_j-orbit not containing the i-edge.

The endpoints b, c are distinct, since ρ_j fixes b by the unique i-crossing lemma but moves c. Take a simple path in M from b to c. Suppose it misses some colour k with $i < k < j$. Its

word factors as uv, where u uses indices less than k and v uses indices greater than k; these factors commute by the string relations. Every generator in v fixes b, since its index is at least $i+2$. Every generator in u fixes c, since its index is at most $j-2$. Thus, using right actions, $c = b^{uv} = b^{vu} = b^u$, while $c^u = c$; applying u^{-1} gives $b = c$, a contradiction. Each intermediate colour must therefore occur. The path has at least $j-i-1$ edges and $j-i$ vertices, proving (2). □

Theorem 3 (The shared bands cannot strictly overlap at high rank)**.** *Use the three regions of Lemma 2, with $j \geq i+2$. Suppose A, D are nontrivial, $G_{<i}$ fixes $M \cup D$, and $G_{>j}$ fixes $A \cup M$. Let*

$$i < h < j, \qquad 0 \leq g < j,$$

where ρ_h acts nontrivially on A and ρ_g acts nontrivially on D. Then

$$n \geq 2r + h - g. \tag{3}$$

In particular, if $r > \lfloor (n-1)/2 \rfloor$, then $g \geq h$, so also $g > i$.

If $g = h$ and the rank bound fails, then $n = 2r$ and

$$|A| = i + h + 1, \quad |M| = j - i, \quad |D| = 2r - j - h - 1. \tag{4}$$

Every simple middle path of minimum length is then Hamiltonian on M and uses each intermediate colour exactly once. If $i > 0$, equality forces $h = i+1$; if $j < r-1$, it forces $h = j-1$. Thus when both crossings are interior the only surviving index configuration is $j = i+2$, $h = g = i+1$, with $|A| = 2i+2$, $|M| = 2$ and $|D| = 2(r-1-j)+2$.

Proof. Apply Lemma 1 to the first crossing. Apply its literal reversal to the second crossing: the reversed indices are $i' = r-1-j$ and $h' = r-1-g$, so

$$|A| \geq i + h + 1, \qquad |D| \geq i' + h' + 1 = 2r - j - g - 1.$$

The other omission orbit is nontrivial in both uses because it contains M and the opposite nontrivial outside region. Adding (2) gives (3). If $g < h$, it gives $n \geq 2r+1$, contradicting failure of the rank target. If $g = h$, failure forces equality everywhere and $n = 2r$. This proves (4), the path assertion, and both strictness consequences from Lemma 1. □

Source routing and attribution. This is a new three-region count for the overlap issue in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.28, physical page 37. The accepted unique-crossing, no-gap, faithful commuting-orbit, active-distance and internal-block arguments, including their retained Whiston input, remain paid dependencies. The source's claim $g > i$ follows here only after the stated orientations and high-rank hypothesis, not just from membership in an active-label set. No minimality or maximality of g, h is needed for the count.

Equality $g = h$ is *not* excluded. Neither are the separate $g > h$ two-orbit equality case, the unproved source support routing, or the complete later graph tables asserted recovered. No new finite computation, formal declaration, original or question credit is claimed.